Modified Ginzburg–Landau Model with MAM

`modified_Ginzburg_Landau_MAM.py` from Zakine 2023 PRX

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Problem

- Consider the following stochastic differential equation (SDE):

$$\frac{\partial \rho}{\partial t} = D\nabla^2\rho + \rho - \rho^3 + \kappa\langle\rho^2\rangle + \sqrt{\epsilon}\,\eta(t), \tag{1}$$

- The noise term $\eta(t)$ is a Gaussian white noise with $\langle\eta(t)\rangle = 0$ and $\langle\eta(t)\eta(t')\rangle = \delta(t-t')$.

From SDE to Freidlin-Wentzell action

- Given a SDE

$$\dot{\rho} = b(\rho) + \sqrt{\epsilon}\sigma(\rho)\eta(t) \tag{2}$$

- The probability of the system to go along a certain path $\rho(x,t)$ is given by

$$P \propto \exp\left(-\frac{1}{\epsilon}S[\rho(x,t)]\right) \tag{3}$$

- The Freidlin-Wentzell action $S[\rho(x,t)]$ is defined as

$$S[\rho(x,t)] = \int_0^{T_{\max}} L(\rho,\dot{\rho})dt, \tag{4}$$

- The Lagrangian $L(\rho,\dot{\rho})$ is given by

$$L(\rho,\dot{\rho}) = \frac{1}{2}\left|\sigma(\rho)^{-1}(\dot{\rho} - b(\rho))\right|^2 = \frac{1}{2}\langle\dot{\rho} - b(\rho), a(\rho)^{-1}(\dot{\rho} - b(\rho))\rangle \tag{5}$$

, where $a(\rho) = \sigma(\rho)\sigma(\rho)^T$.

From Lagrangian to Hamiltonian

- The Hamiltonian $H(\rho, \theta)$ is given by Legendre transformation:

$$H(\rho, \theta) = \sup_{\dot{\rho}} \left[\langle \theta, \dot{\rho} \rangle - L(\rho, \dot{\rho}) \right] \tag{6}$$

  $\langle \theta, \dot{\rho} \rangle$ is the inner product over the space of x.
- To find the optimal $\dot{\rho}$, we take the derivative of $H(\rho, \theta)$ with respect to $\dot{\rho}$ and set it to zero:

$$\frac{\partial}{\partial \dot{\rho}} \left[\langle \dot{\rho}, \theta \rangle - \frac{1}{2} \langle \dot{\rho} - b(\rho), a(\rho)^{-1} (\dot{\rho} - b(\rho)) \rangle \right] = 0 \tag{7}$$

- Solving for $\dot{\rho}$, we get

$$\dot{\rho}^* = b(\rho) + a(\rho)\theta \tag{8}$$

- Substituting $\dot{\rho}^*$ into $H(\rho, \theta)$, we get

$$H(\rho, \theta) = \langle b(\rho), \theta \rangle + \frac{1}{2} |\sigma(\rho)\theta|^2 \tag{9}$$

Modified Ginzburg-Landau Model

- Let $b(\rho) = D\nabla^2\rho + \rho - \rho^3 + \kappa\langle\rho^2\rangle$ and $\sigma(\rho) = 1$.
- The Hamiltonian $H(\rho, \theta)$ is given by

$$H(\rho, \theta) = \langle D\nabla^2\rho + \rho - \rho^3 + \kappa\langle\rho^2\rangle, \theta\rangle + \frac{1}{2}\|\theta\|^2 \tag{10}$$

- Then the Lagrangian $L(\rho, \dot{\rho})$ is defined as purely deterministic cost:

$$L(\rho, \dot{\rho}) = \frac{1}{2}\|\dot{\rho} - b(\rho)\|^2 = \frac{1}{2}\langle\dot{\rho} - b(\rho), \dot{\rho} - b(\rho)\rangle \tag{11}$$

- *Physical meaning:* This Lagrangian directly corresponds to the required noise energy $\frac{1}{2}\epsilon\|\eta(t)\|^2$ to drive the trajectory.

Minimum Action Method (MAM)

- Solves the **minimum action** (instanton) path for a modified Ginzburg–Landau model in a spatially discrete setting.
- Uses a **Hamiltonian formulation** with state ρ (density/order parameter) and conjugate momentum θ (noise force).
- Time evolution is in **minimization time** τ; the path in "time" $t \in [0, T_{\max}]$ is discretized with N_t points.

Discretize Hamiltonian and Lagrangian

For spacing h, the Hamiltonian (sum over boxes with total length $L_x = N_x \times h$) is

$$H(t;\rho,\theta) = h\sum_{i=1}^{N_x}\left[(D\nabla^2\rho_i + \rho_i - \rho_i^3 + \kappa\langle\rho^2\rangle)\theta_i + \frac{1}{2}\theta_i^2\right]. \tag{12}$$

where ρ_i, θ_i are the density and noise force at the i-th box, $\nabla^2\rho_i = \frac{\rho_{i+1}+\rho_{i-1}-2\rho_i}{h^2}$ is the discrete Laplacian, $\langle\rho^2\rangle = \frac{1}{N_x}\sum_{i=1}^{N_x}\rho_i^2$ is the spatial average of the square of the density.
Lagrangian (along the path in t):

$$L(t;\rho,\dot{\rho}) = h\sum_{i=1}^{N_x}[\dot{\rho}_i\cdot\theta_i] - H(\rho,\theta). \tag{13}$$

Boundary conditions and initial path

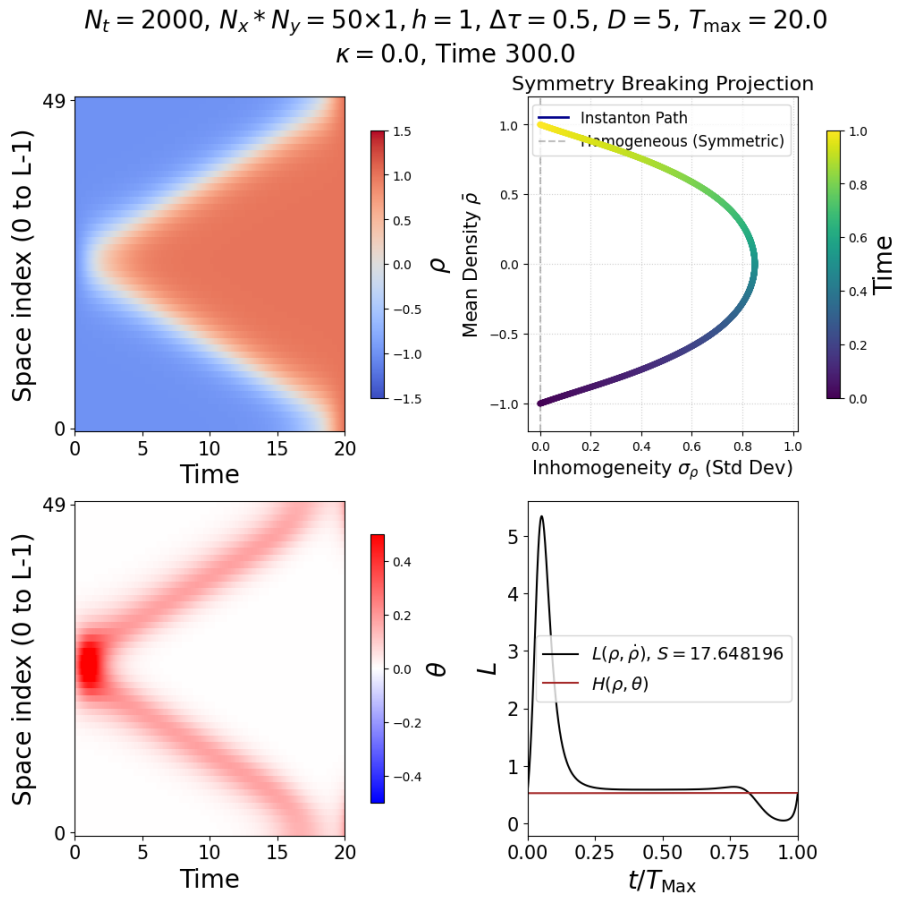
- We use the method of Zakine and vanden Eijnden to obtain solutions to Hamilton's equations with boundary conditions given by the endpoints ρ_1, ρ_2.
- ρ_1, ρ_2 are homogeneous (all boxes equal) from the roots of the cubic equation: $-x^3 + \kappa x^2 + x = 0$; for `upward=True`, path goes from smallest to largest root.
- Initial path: For $0 \leq j \leq N_x - 1$ and $1 \leq k \leq N_t - 2$,

$$\rho_j^k = (1 - k/N_t)\,\rho_1 + k/N_t\,\rho_2 \tag{14}$$

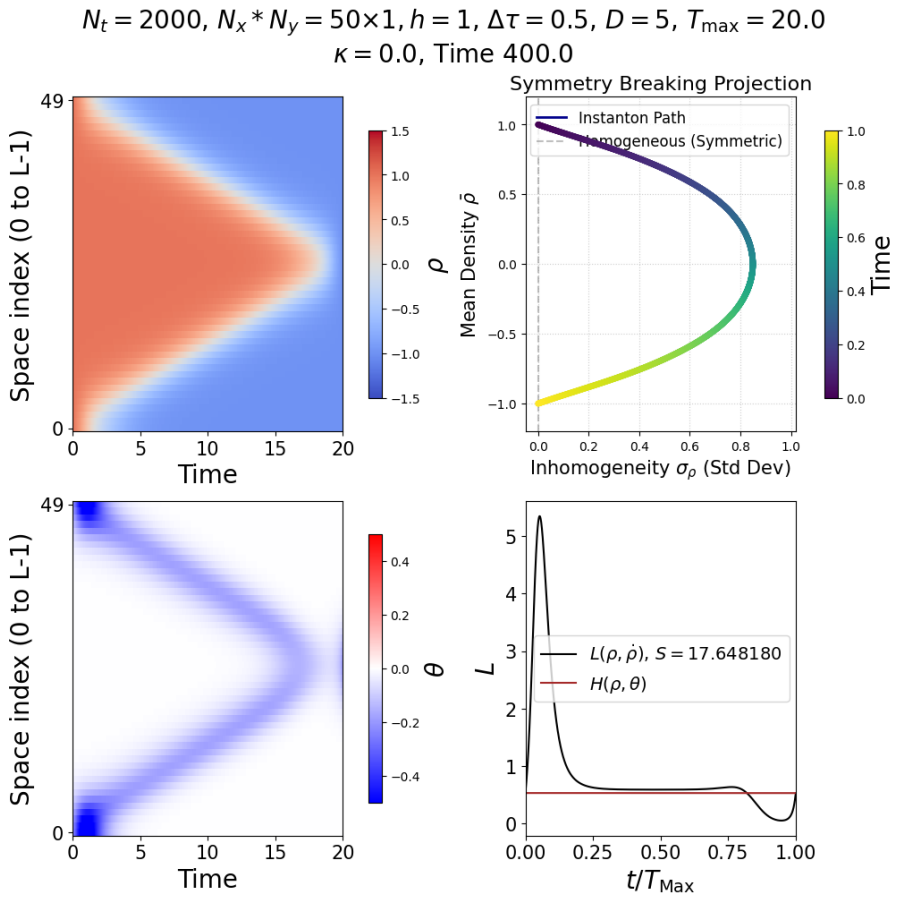
Visualization

- **Heatmaps:** ρ and θ as functions of (time along path, space index).
- **Symmetry-breaking projection:** Scatter of mean density $\bar{\rho}$ vs. standard deviation σ_ρ along the path (homogeneous vs. symmetry-broken).
- **L & H:** Lagrangian L and Hamiltonian H vs. t/T_{max}.

Example run: when $\kappa = 0$

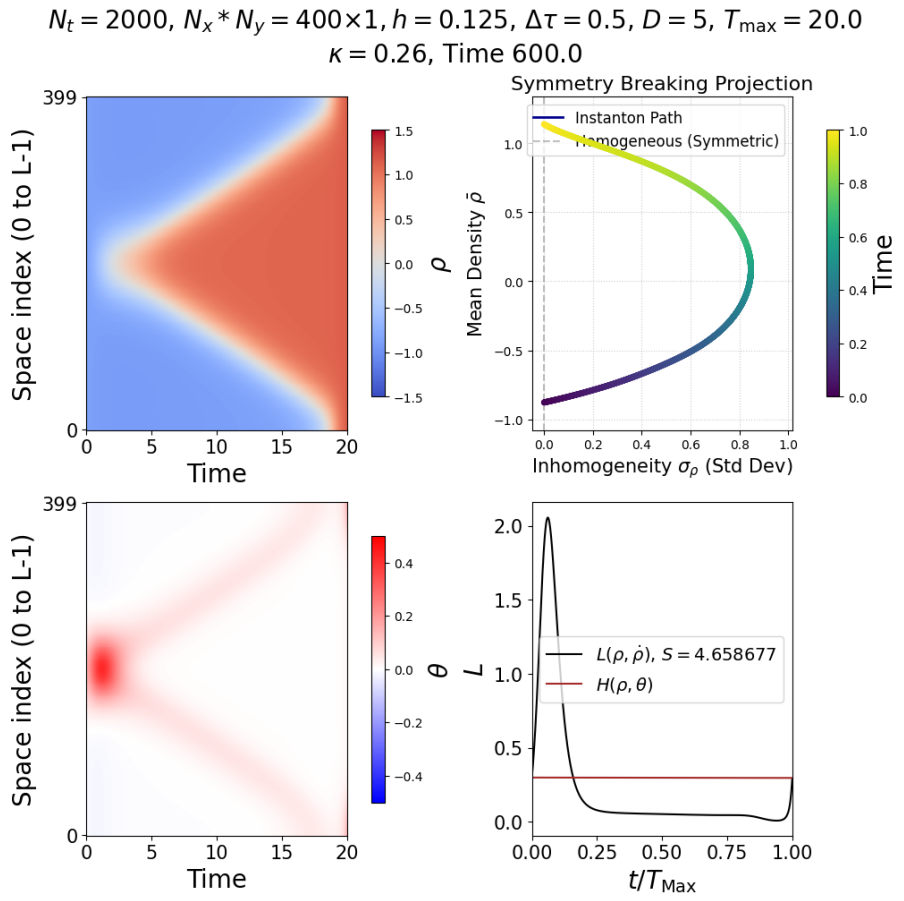


Forward

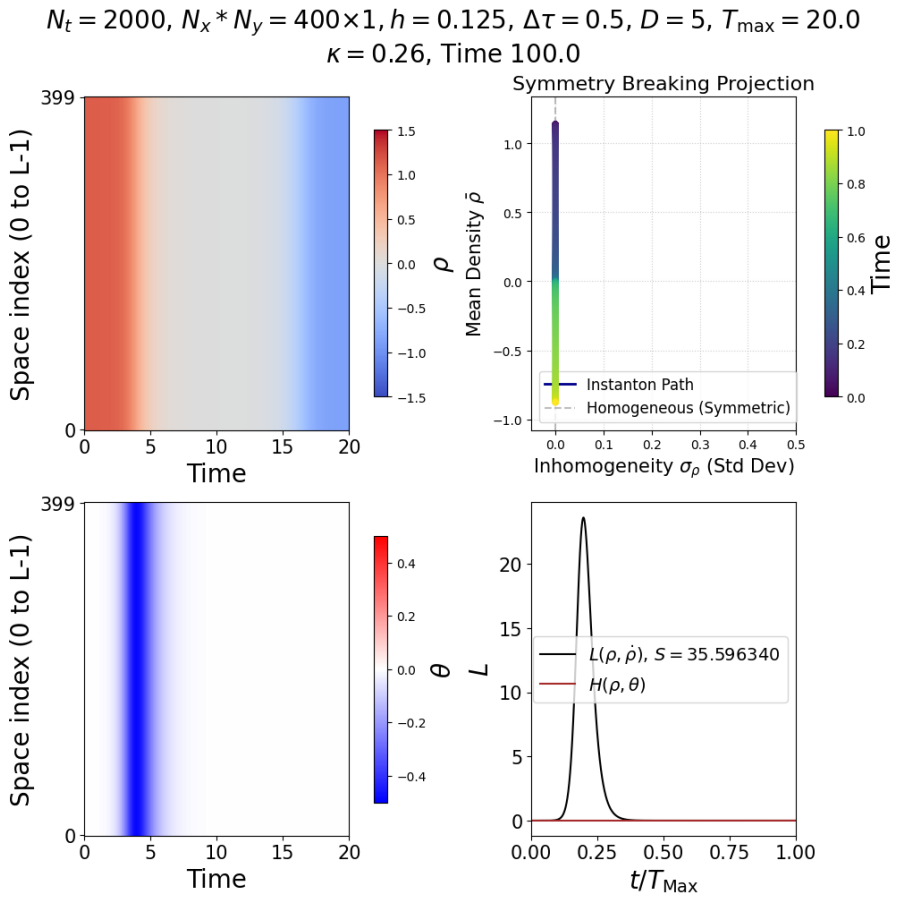


Backward

Example run: when $\kappa \neq 0$

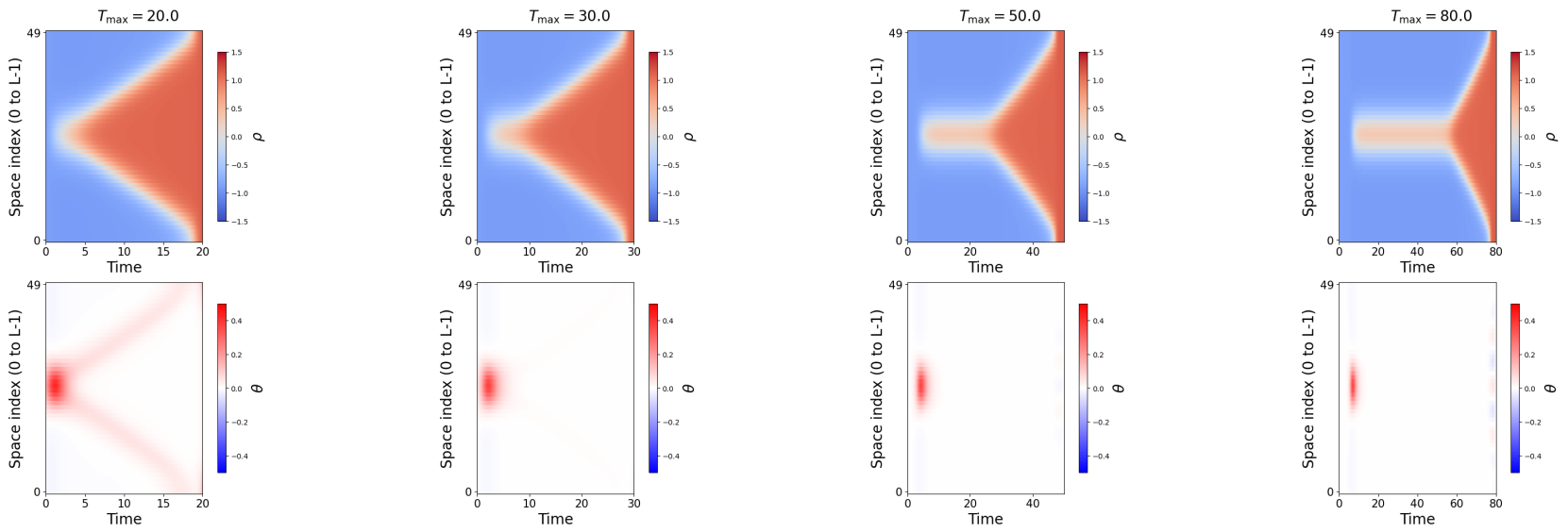


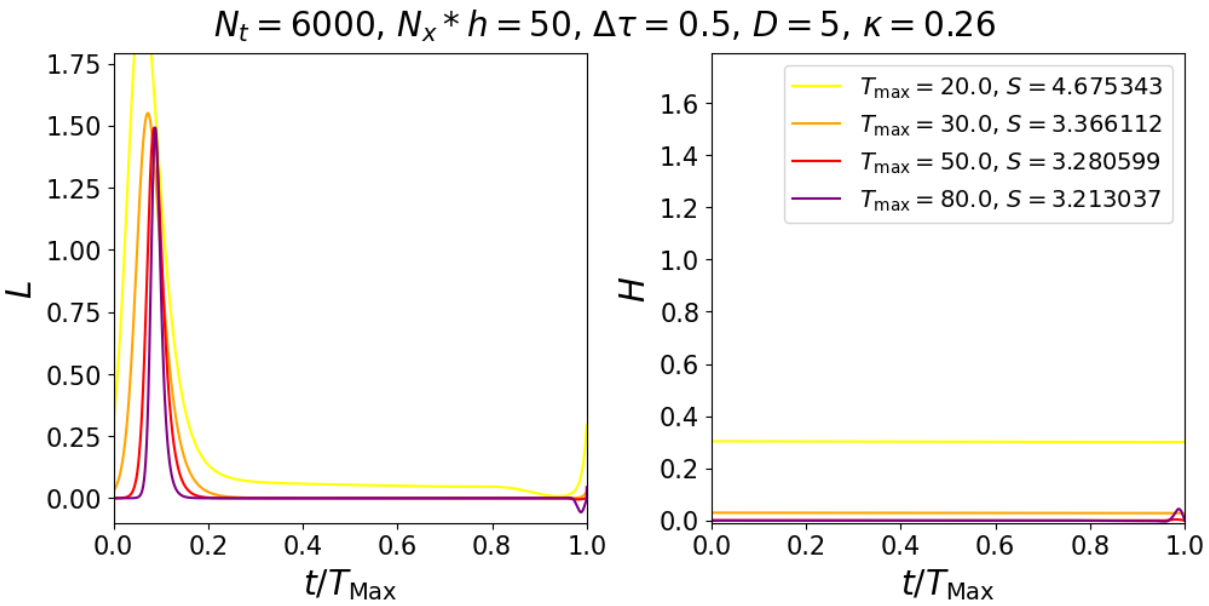
Forward



Backward

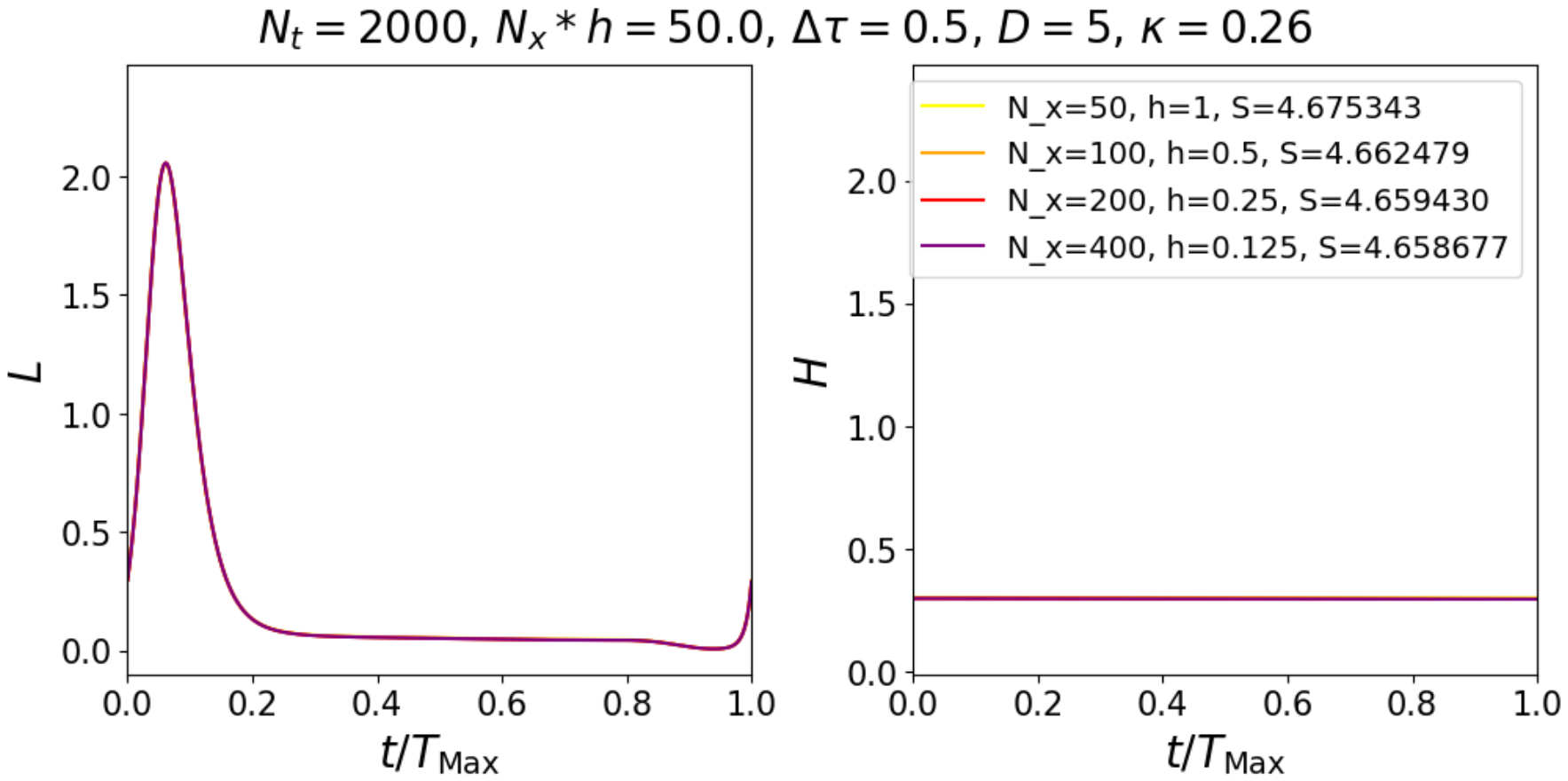
Varying T_{max} (forward path)





Instanton of different κ

Results are independent of spatial resolution



Results are independent of temporal resolution

Summary

- `modified_Ginzburg_Landau_MAM.py` implements a **minimum-action path** for a modified Ginzburg–Landau model on L spatial boxes.
- Uses (ρ, θ) and (U, V) formulations, FFT in the path direction, and semi-implicit tridiagonal solves per Fourier mode.
- Output: action S, heatmaps of ρ/θ, and a symmetry-breaking plot.

Discretization: U and V variables

The code uses the change of variables

$$U = \rho + \theta, \qquad V = \rho - \theta \quad \Rightarrow \quad \rho = \frac{U+V}{2}, \quad \theta = \frac{U-V}{2}. \tag{15}$$

$$\partial_\tau U = \partial_t U + f(U,V) \quad \Rightarrow \quad \partial_\tau V = -\partial_t V + g(U,V) \tag{16}$$

- Characteristic lines for U and V are $d\tau = -dt$, $d\tau = dt$ respectively.
- In other words, we update U backward in t and V forward in t when iterating τ.

This yields a semi-implicit scheme that is solved in Fourier space (over the N_{copy} index) using prebuilt tridiagonal matrices `A_solve_upper_adapted`, `B_solve_lower_adapted` per Fourier mode.

Implicit upwind scheme: why "upwind"?

Challenge: How do we discretize the grid so we capture both waves without numerical explosion? **Central difference** uses both neighbors $\Rightarrow$ wrong direction for a wave $\Rightarrow$ checkerboard / instability. **Upwind:** "Where the wind comes from, take the difference from there."

- V **(forward in** t**):** information arrives from the *past* $(j-1)$. Use **backward difference** for $\partial_t V$: $(V_j^{k+1} - V_{j-1}^{k+1})/\Delta t$.
- U **(backward in** t**):** information arrives from the *future* $(j+1)$. Use **forward difference** for $\partial_t U$: $(U_{j+1}^{k+1} - U_j^{k+1})/\Delta t$.

So the stencil respects the direction of each characteristic.

Why "implicit"? Unconditional stability

Upwind alone can still blow up when $\Delta\tau$ is large. **Implicit** trick: evaluate the convective derivative at the *future* time $k+1$ (unknowns on the left). Then the scheme is **unconditionally stable**. Let $r = \Delta\tau/\Delta t$ (path step Δt). **Wave** V ($\partial_\tau V = -\partial_t V + g$): backward difference in t at $k+1$:

$$\frac{V_j^{k+1} - V_j^k}{\Delta\tau} = -\frac{V_j^{k+1} - V_{j-1}^{k+1}}{\Delta t} + g_j^k \;\Rightarrow\; V_j^{k+1}(1+r) - V_{j-1}^{k+1}\, r = V_j^k + \Delta\tau\, g_j^k.$$

Wave U ($\partial_\tau U = \partial_t U + f$): forward difference in t at $k+1$:

$$\frac{U_j^{k+1} - U_j^k}{\Delta\tau} = \frac{U_{j+1}^{k+1} - U_j^{k+1}}{\Delta t} + f_j^k \;\Rightarrow\; U_j^{k+1}(1+r) - U_{j+1}^{k+1}\, r = U_j^k + \Delta\tau\, f_j^k.$$

These yield a **lower** triangular system for V and an **upper** triangular system for U.

Why lower triangular? Macro view of the V-update

At one grid point: $v_j^{k+1}(1+r) - v_{j-1}^{k+1} r = \mathrm{RHS}_j$. To see the matrix shape, write **all** equations $j = 1, \ldots, N$ as $A\mathbf{v} = \mathbf{b}$. **Unknown:** $\mathbf{v}^{k+1} = (v_1^{k+1}, \ldots, v_N^{k+1})^\top$; $\mathrm{RHS}_j = v_j^k + \Delta\tau\, g_j^k$ (known). Boundary v_0^{k+1} is given $\Rightarrow$ move to RHS. Row $j = 1$: $(1+r)v_1^{k+1} = \mathrm{RHS}_1 + r v_0^{k+1}$. Row $j \geq 2$: $-r v_{j-1}^{k+1} + (1+r) v_j^{k+1} = \mathrm{RHS}_j$.

$$\begin{bmatrix} 1+r & 0 & 0 & \cdots \\ -r & 1+r & 0 & \cdots \\ 0 & -r & 1+r & \cdots \\ \vdots & \vdots & \ddots & \ddots \end{bmatrix} \begin{bmatrix} v_1^{k+1} \\ v_2^{k+1} \\ \vdots \\ v_N^{k+1} \end{bmatrix} = \begin{bmatrix} \mathrm{RHS}_1 + r v_0^{k+1} \\ \mathrm{RHS}_2 \\ \vdots \\ \mathrm{RHS}_N \end{bmatrix}$$

So the coefficient matrix is **lower triangular** (diagonal $(1+r)$, subdiagonal $-r$, above diagonal 0). Hence **forward substitution**: solve v_1, then v_2, ..., v_N—physically, V propagates *forward* in t. U-**update:** $u_j^{k+1}(1+r) - u_{j+1}^{k+1} r = \mathrm{RHS}_j$ depends on j and $j+1$ $\Rightarrow$ **upper triangular** matrix $\Rightarrow$ **backward substitution** from $j = N$ down to 1 (backward in t).

Time stepping

Concept (each $\Delta\tau$):

1. **Downstream:** Solve V (lower triangular, forward substitution) from $t = 0$ to $T_{\max}$ with boundary V_0^{k+1}.
2. **Upstream:** Solve U (upper triangular, backward substitution) from $t = T_{\max}$ back to 0 with boundary at last copy.
3. **Update:** $\rho = \frac{U+V}{2}$, $\theta = \frac{U-V}{2}$. Code: `A_solve_upper_adapted` $\Rightarrow$ U; `B_solve_lower_adapted` $\Rightarrow$ V.

Implementation: (1) Compute $\partial H/\partial\rho$, $\partial H/\partial\theta$; form `reaction_U` ($f(u, v)$), `reaction_V`($g(u, v)$); FFT in copy direction. (2) `solve_triangular(A_..., ...)` for U (BC at last copy $\rho = \rho_2$); `solve_triangular(B_..., ..., lower=True)` for V (BC at first copy $\rho = \rho_1$). (3) FFT^{-1}, update ρ, θ, re-set U, V. Boundaries keep $\rho(0) = \rho_1$, $\rho(N_{\text{copy}} - 1) = \rho_2$.

Connecting boundary conditions of U and V

We know *both* endpoints of ρ: $\rho(0) = \rho_1$ and $\rho(T_{\max}) = \rho_2$. From $U = \rho + \theta$ and $V = \rho - \theta$ we have $U + V = 2\rho$. At the end of the path:

$$U(T_{\max}) = 2\rho_2 - V(T_{\max}). \tag{17}$$

We use the *old* V from the previous iteration and the fixed ρ_2 to get U at the endpoint; then we propagate U *backward* in t down to $t = 0$. At the start of the path:

$$V(0) = 2\rho_1 - U(0). \tag{18}$$

U has just finished its "time travel" and we have the updated $U(0)$. We use it with the fixed ρ_1 to get V at the start; then we propagate V *forward* in t up to $T_{\max}$.